Explanatory and Predictive Modelling of E-Commerce Order Revenue

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0.1 Introduction

SweetAroma is a British high-end online perfume business aiming to deepen its understanding of customer spending behaviour as the business continues to expand.

This report addresses two key business questions. First, it analyses how the provided factors influence order revenue on the SweetAroma website. Second, it develops a predictive model capable of estimating the revenue of future orders based on these characteristics, enabling the company to better attract high-value customers.

0.2 Data Understanding

To support this analysis, SweetAroma provided two datasets. The main dataset contains 10,000 rows of orders placed on the SweetAroma website in July 2025, and 6 columns of variables, summarised in Table 1.

Variable	**Description**	**Type**
revenue	Revenue (£) generated from the order (before any cost or discount).	Numerical
ad_channel	Advertising source that led the customer to the website (1 = No Ad, 2 = Paid Search Ad, 3 = Free SEO, 4 = Online Ad).	Categorical
number_past_order	Number of previous orders made by the customer.	Numerical
past_spend	Total historic spending (£) by the customer before this order.	Numerical
time_web	Time spent on the website (seconds) before placing the order.	Numerical
voucher	Whether a 5% discount was used (0 = No, 1 = Yes).	Categorical

Table 1: Description of variables in the July 2025 dataset.

In this analysis, *revenue* serves as the target variable, while the remaining five variables act as predictors. The second dataset provided contains 20 new orders with the same five predictor variables; however, the target variable (*revenue*) has been withheld. Instead, this dataset includes a categorical *order* column, which simply numbers each observation from 1 to 20. This dataset will be used solely to test the predictive model. Table 2 displays summary statistics of the continuous variables in the main dataset.

Variable	**Mean**	**Standard Deviation**
revenue	55.46	13.28
time_web	122.35	24.26
past_spend	10.10	13.24

Table 2: Mean and standard deviations of the *revenue*, *time_web* and *past_spend* variables.

0.3 Data Preparation

The main dataset contained 581 rows with missing values (5.81%). Little's MCAR test returned a p-value of 0.415, indicating that the data is likely MCAR (missing completely at random).

The bar plot in Figure 1 shows that the five predictor variables have similar levels of missingness, ranging from 103 to 134 cases, while the target variable *revenue* has none. The combinations plot indicates very few instances of multiple variables missing simultaneously, suggesting that missingness is independent across variables and therefore likely MCAR. As a result, listwise deletion was applied, as it is unlikely to introduce bias.

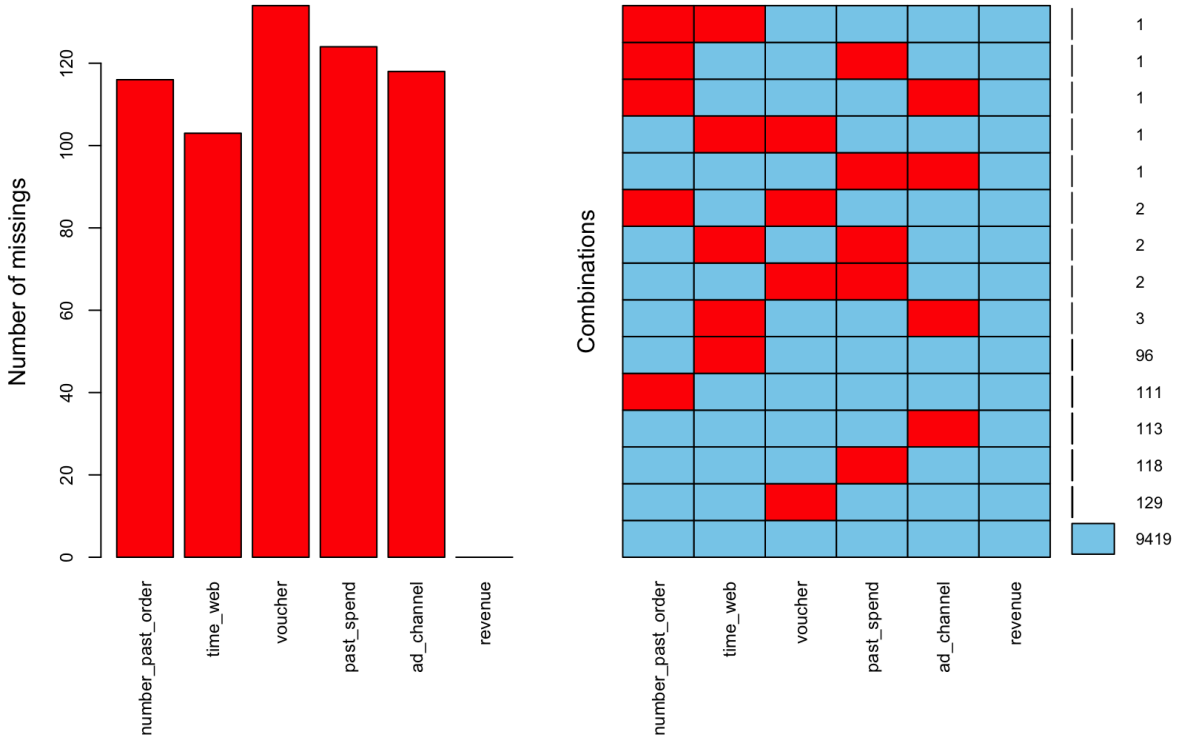


Figure 1: Bar and combinations plot of missing values in July 2025 dataset.

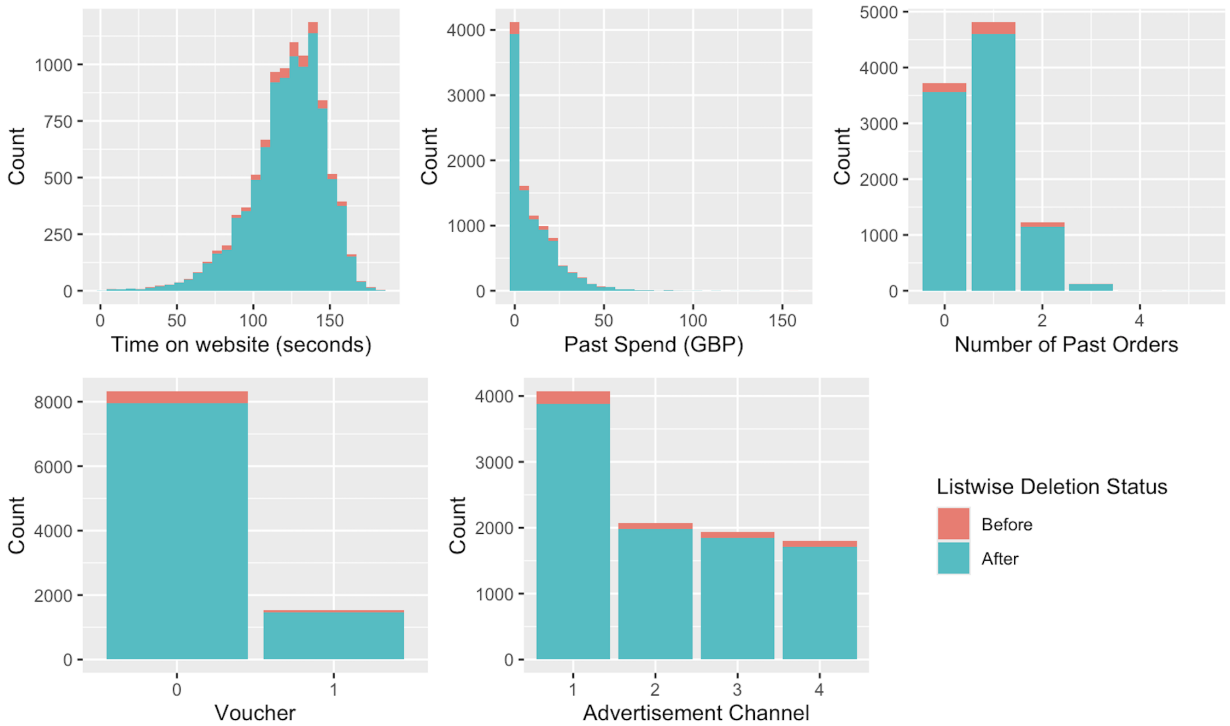


Figure 2: Frequency histograms (*time_web* and *past_spend*) and bar charts (*number_past_order*, *voucher* and *ad_channel*) before and after listwise deletion.

In Figure 2, the frequency distributions for all variables appear largely unchanged after applying listwise deletion. Furthermore, the revised mean values for *time_web* and *past_spend* (122.42 and 10.09), along with their standard deviations (24.19 and 13.16), show minimal change compared with the pre-deletion statistics in Table 2. This suggests that removing rows with missing values has not significantly altered the underlying distributions, indicating that listwise deletion is unlikely to introduce bias in this analysis.

0.4 Modelling

Multiple linear regression was selected for this analysis because it allows us to quantify the relationship between the several independent variables and continuous dependent variable, while also providing a predictive model for estimating revenue.

0.4.1 Dummification

To implement linear regression, categorical variables must be converted into dummy variables. In the July 2025 dataset, the *voucher* variable was already in the correct binary format, requiring no modification. However, the *ad_channel* variable was one-hot encoded into three dummy variables corresponding to the final three of the four advertising source: 1 if the customer arrived via that channel and 0 otherwise. A fourth dummy variable was unnecessary, as the first advertising source is implicitly represented when all three dummy variables equal 0.

0.4.2 Correlation

To initially investigate the relationship between numerical variables, correlation coefficients were calculated. Figure 3 visualises these correlations.

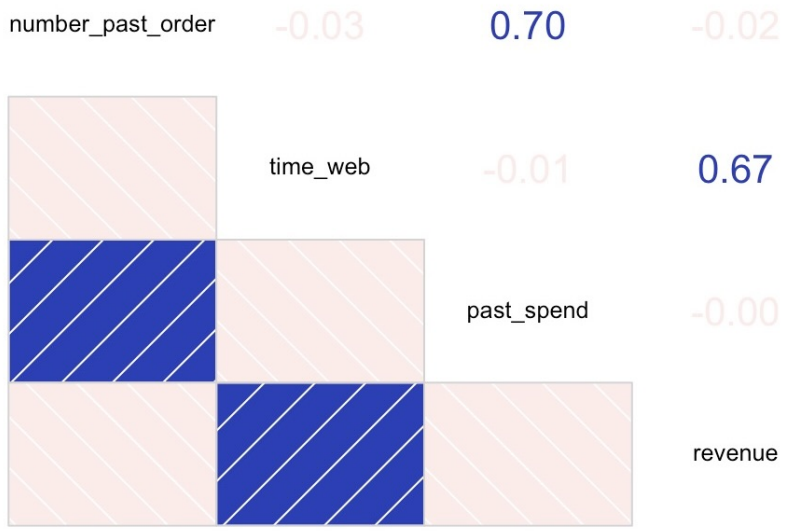


Figure 3: Correlation matrix of numerical variables, with coefficients in the upper panels and a colour scale in the lower panels (blue = positive, red = negative, darker = stronger).

The correlation matrix shows that the only numerical variable with a significant correlation to the target variable, *revenue*, is the time spent on the website, indicating that *time_web* may be an important

factor in customer spending on SweetAroma, and also may be a feature of high importance in predictive modelling. Furthermore, *number_past_order* and *past_spend* exhibit a strong positive correlation coefficient of 0.7, indicating multicollinearity. This may suggest that these variables will work against eachother in predictive modelling.

0.4.3 Checking Assumptions

To assess the robustness of the linear regression, three assumptions were checked:

1. A linear relationship between the predictors and the dependent variable.
2. Constant variance of the dependent variable across predictor values.
3. Independence of the residuals.

Initial assumption checks were conducted using the visualisations in Figure 4.

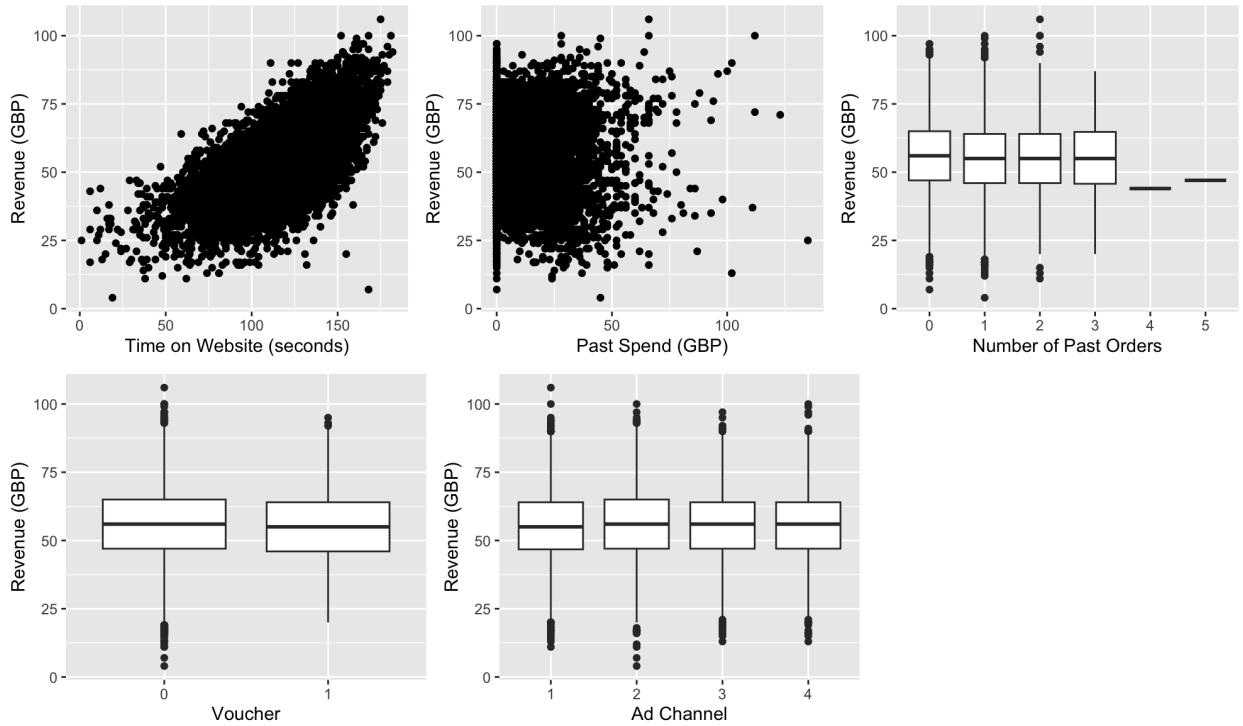


Figure 4: Scatterplots of *revenue* against *time_web* and *past_spend*, and boxplots of *revenue* by *number_past_order*, *voucher* and *ad_channel*.

Time on website exhibited a clear linear relationship with revenue, while past spend and number of past orders displayed weaker linearity; variance appeared broadly stable for most predictors, though past spend showed some heteroscedasticity; no visible autocorrelation was present. Overall, the assumptions were sufficiently met to proceed with linear regression, while noting that some predictors may require cautious interpretation.

0.4.4 Regression for Explanation

To identify how customer factors influence spending and address the first business question, a linear regression model was built in R using all available predictors. Table 3 summarises the resulting coefficients

for each variable.

	Estimate	Standard Error	t value	Pr(>|t|)
number_past_order	-0.3981	0.2033	1.9570	0.0504
time_web	0.3659	0.0042	86.6051	<2e-16
voucher	-0.7570	0.2826	-2.6791	0.0074
past_spend	0.0242	0.0109	2.2244	0.0262
ad_channel_2	-0.0285	0.2737	-0.1043	0.9170
ad_channel_3	-0.0875	0.2805	-0.3119	0.7552
ad_channel_4	-0.2277	0.2880	-0.7909	0.4291

Table 3: Coefficient estimates, standard errors, t-values, and p-values for each predictor variable.

0.4.5 Regression for Prediction

To build a predictive model, variables that were not statistically significant were removed using the backwards elimination method of stepwise regression.

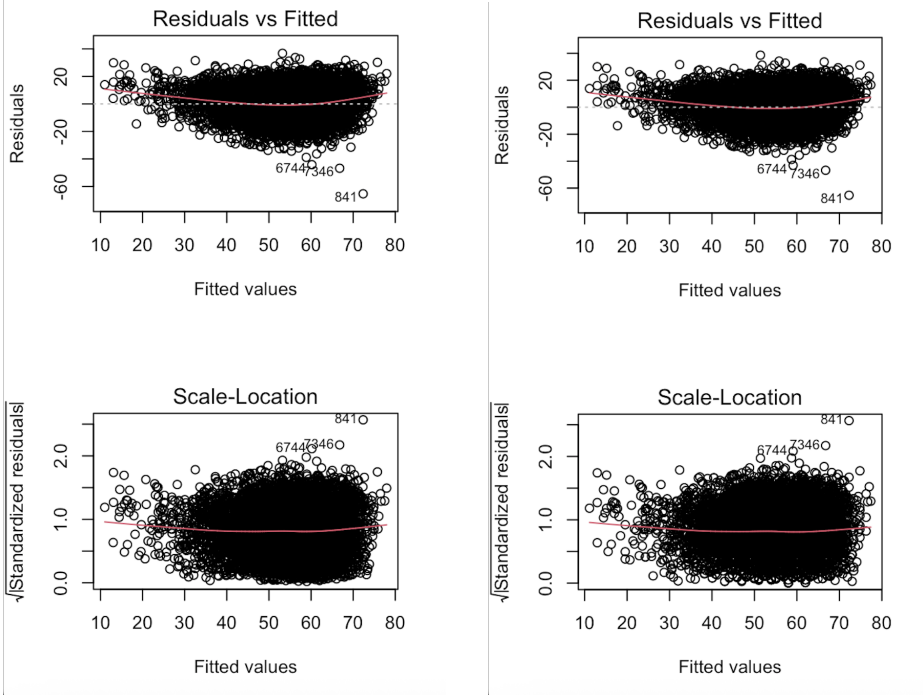
The *ad_channel* dummies and *number_past_order* were removed first, followed by *past_spend*, which became insignificant once its correlated counterpart was excluded. The final predictive model therefore retained only *time_web* and *voucher*, both of which remained statistically significant. From this model, the revenues of the orders in the new customer dataset were predicted, as displayed in Table 4.

Order	Prediction
1	64.22
2	62.73
3	47.75
4	55.07
5	46.29
6	58.00
7	64.22
8	56.54
9	60.54
10	65.32
11	61.27
12	47.02
13	52.14
14	16.27
15	59.83
16	66.79
17	43.72
18	51.41
19	62.03
20	64.95

Table 4: Revenue predictions for orders 1-20 from the new customer dataset provided by SweetAroma.

0.5 Evaluation

To evaluate the models, their robustness was first assessed by returning to the assumptions for linear regression.



(a) Explanatory model. (b) Predictive model.

Figure 5: Residuals vs fitted values (top) and residuals vs standardised residuals (bottom) scatter plots.

Figure 5 shows little evidence of heteroscedasticity in either model: the red trend lines remain approximately flat, and the residual spread is fairly constant across fitted values, supporting Assumption 2. Assumption 3 was assessed using the Durbin–Watson test. The p-values for the explanatory and predictive models (0.1113 and 0.1154) indicated no evidence of autocorrelation at the 5% level, so this assumption was satisfied.

To address the first business question, the significance of each factor was examined. Table 3 shows that *ad_channel* and *number_past_order* are not statistically significantly related to revenue, with p-values greater than 0.05. In contrast, *time_web* has a very large t-value, indicating a strong and highly significant positive relationship with revenue: each additional second spent on the website increases expected revenue by approximately £0.3659, with a very small standard error and an effectively zero probability of this effect occurring by chance. Voucher usage is also statistically significant, with the use of a voucher associated with an average reduction of £0.7570 in revenue. Finally, *past_spend* has a smaller but statistically significant positive effect, with an increase of £1 increasing expected revenue by £0.0242.

Model fit was assessed using R^2 and adjusted R^2. The explanatory model achieves values of 0.4442 and 0.4438, explaining roughly 44% of the variation in revenue. The predictive model performs similarly (0.4438 and 0.4437), indicating moderate predictive accuracy. Its performance will be further evaluated using the root mean square error against the actual revenue of the 20 new customers.

0.6 Conclusion

This analysis demonstrates that the time spent on the SweetAroma website is the factor with the most significant relationship to revenue, with longer times associated with higher spending. Voucher use and a customer's total past spend are also statistically significant in the explanatory model: vouchers reduce revenue, while past spend increases it. However, the significance of *past_spend* appears to be influenced by its strong positive correlation with *number_past_order*. The remaining factors (*ad_channel*, and *number_past_order*) do not significantly influence revenue. Overall, the explanatory model accounts for around 44% of variation in revenue.

The final predictive model retained only the statistically significant variables *time_web* and *voucher*. Although the model's R^2 and adjusted R^2 values of 0.44 only indicate moderate explanatory power, its statistically significant predictors still provide SweetAroma with meaningful insights into customer behaviour that can support targeting high-value customers.